DeepWatch: A Unified Deepfake Forensics and Attribution Platform

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Executive Summary

Anyone can now create a fake face, voice, or video in minutes — and real victims are losing money to it. Today's research tools each solve only one piece: one checks if a face is fake, another names the tool that made it, a third checks only voice. None work together, none explain their answer, none produce proof a victim can use. This report studies four IEEE/CVF papers, explains what each cannot do, and proposes **DeepWatch**: one platform that detects fakes, names the tool that made them, shows the evidence, and prints a forensic report.

1. The Real-World Problem

A parent gets a video call from "their child" asking for emergency money — the face and voice are AI-generated. A victim finds a fake video of themselves online and cannot prove it is fake. Deepfake scams cost victims billions globally, and the technology improves every month. The people most harmed are ordinary users with no forensic tools and no way to *prove* a fake when it matters. That is the problem this project addresses.

2. Research Objectives

The objectives are to (1) study how the best published systems detect and attribute deepfakes; (2) identify precisely what each cannot do; and (3) design one integrated platform that fixes those weaknesses, with improvements measurable against the base papers' published numbers.

3. The IEEE Base Papers and Their Disadvantages

Four verified IEEE/CVF papers form the technical foundation. Each is strong, but each leaves a clear gap.

Base paper & link	What it does	Its disadvantage (plain terms)
UCF — Yan et al., ICCV 2023 [1] arxiv.org/abs/2304.13949	Detects fake faces via "common" forgery signs shared by all methods. 0.996 AUC on familiar data.	Drops to 0.828 on unfamiliar fakes; needs costly per-method labels; gives a bare score, no explanation.
Up-sampling artifacts — Tan et al., CVPR 2024 [2] doi.org/10.1109/CVPR52733.2024.02657	Catches fakes via fingerprints older AI generators leave when enlarging images.	Blind to new "diffusion" generators — they leave no such fingerprint. Works only on yesterday's fakes.
OW-DFA / CPL — Sun et al., ICCV 2023 [3] arxiv.org/abs/2309.11132	Names *which* forgery method made a fake — even unseen ones.	Faces only — no voices or scenes. Low accuracy on unseen methods; field is young.
ASVspoof 2021 — Liu et al., IEEE/ACM TASLP 2023 [4] arxiv.org/abs/2210.02437	Standard benchmark for detecting fake **speech**.	Audio only — never fused with video, though a fake call has both fake face and fake voice.

4. What We Will Newly Implement

DeepWatch combines the strengths above and fixes each listed disadvantage:

- **Diffusion-aware detection** — catch the newest generators paper [2] misses, using frequency/noise-pattern analysis instead of up-sampling fingerprints alone.
- **Audio + video fusion** — check face, voice, and lip-sync together in one pipeline; a real-looking video with synthetic sound is still caught (merging [1] and [4]).
- **Beyond-face attribution** — extend [3]'s open-world approach to name the likely generation tool across more forgery types.
- **Explainable evidence** — heatmaps and plain-language reasons ("unnaturally blurred ear region") instead of a bare score.
- **One-click forensic report** — a tamper-evident PDF for police or platform moderators; no reviewed system produces usable evidence.

5. Why This Project Stands Out

Most student projects re-implement a published result. DeepWatch differs in three ways. **First**, deepfake fraud is the defining technology issue of this era — it is in the news weekly and understood instantly by any audience. **Second**, the novelty is structural, not cosmetic — detection, attribution, and audio analysis exist as three separate fields; uniting them into one forensic pipeline is something no reviewed paper does. **Third**, the improvement is provable — the base papers publish exact numbers (0.828 AUC, attribution error floors), so "we did better" can be demonstrated, not just claimed.

6. Our Reach and Expected Impact

The project is designed for real use, not only a lab demo. Beneficiaries: ordinary citizens verifying suspicious calls or media, small police cyber cells without forensic tools, journalists checking viral content, and awareness programs. Technically, the work contributes (a) an evaluated audio–video fusion method, (b) a compression-robust evaluation on realistic social-media-quality media, and (c) an open benchmark against the base papers' published results. The deliverable is a working web platform: upload media → get detection score, attribution, evidence heatmaps, and a forensic report in under a minute.

7. Challenges and Conclusion

Cross-dataset generalization is the field's hardest open problem, so improvement targets will be honest and modest; compression and audio–video synchronization add further engineering demands. Conclusion: deepfake forensics is a real, current, under-served problem. The four IEEE/CVF base papers provide proven components but each leaves a documented gap — poor generalization, blindness to new generators, face-only attribution, and audio–video separation. DeepWatch closes these gaps in one usable platform with provable improvements, and is therefore selected as the project direction.

References

[1] Z. Yan, Y. Zhang, Y. Fan, B. Wu, "UCF: Uncovering Common Features for Generalizable Deepfake Detection," IEEE/CVF ICCV 2023. DOI: 10.1109/ICCV51070.2023.02048 — https://arxiv.org/abs/2304.13949

[2] C. Tan et al., "Rethinking the Up-Sampling Operations in CNN-Based Generative Network for Generalizable Deepfake Detection," IEEE/CVF CVPR 2024. DOI: 10.1109/CVPR52733.2024.02657 — doi.org/10.1109/CVPR52733.2024.02657

[3] Z. Sun et al., "Contrastive Pseudo Learning for Open-World DeepFake Attribution," IEEE/CVF ICCV 2023 — https://arxiv.org/abs/2309.11132

[4] X. Liu et al., "ASVspoof 2021: Towards Spoofed and Deepfake Speech Detection in the Wild," IEEE/ACM Trans. Audio, Speech, and Language Processing, 2023. DOI: 10.1109/TASLP.2023.3285283 — https://arxiv.org/abs/2210.02437